MODULE 1

Inventors at Work

"I will not follow where the path may lead, but I will go where there is no path, and I will leave a trail."

— MURIEL STRODE

ESSENTIAL QUESTION

What kinds of circumstances push people to create new inventions?

MAIN SELECTIONS

1	**The Inventor's Secret: What Thomas Edison Told Henry Ford**	· Suzanne Slade (illus. Jennifer Black Reinhardt)	NARRATIVE NONFICTION
2	**Winds of Hope**	· Katy Duffield	NARRATIVE NONFICTION
3	**Wheelchair Sports: Hang Glider to Wheeler-Dealer**	· Simon Shapiro (art by Theo Krynauw & Warwick Goldswain)	INFORMATIONAL TEXT
4	**Captain Arsenio: Inventions and (Mis)Adventures in Flight**	· Pablo Bernasconi	SCIENCE FICTION / FANTASY

CONTENTS

A · Vocabulary (34) ..... p. 2

B · Reading Comprehension (32) ..... p. 6

C · IB Critical Thinking (5) ..... p. 12

D · MAP Practice (20) ..... p. 15

Worksheet A — Vocabulary

Module 1 · Inventors at Work · Grade 5 · 34 Items

CCSS L.5.4 · L.5.5 · L.5.6

NAME _____ DATE _____

Directions. Read each question carefully. Watch out for tricky choices. Fill in the circle next to the BEST answer.

BIG IDEA WORDS

Q1 L.5.6

excel

If you **excel** at something, you —

- ◯ **A.** are afraid to do it.
- ◯ **B.** have never tried it before.
- ◯ **C.** are very good at it.
- ◯ **D.** do it only by accident.

Q2 L.5.6

transcend

If you **transcend** a boundary, you —

- ◯ **A.** stop right in front of it.
- ◯ **B.** draw a picture of it.
- ◯ **C.** go above or beyond it.
- ◯ **D.** hide far away from it.

Q3 L.5.6

illustrious

An **illustrious** person is —

- ◯ **A.** very tall and strong.
- ◯ **B.** famous for his or her achievements.
- ◯ **C.** brand new to a job.
- ◯ **D.** drawn with bright colors.

Q4 L.5.6

revere

If you **revere** someone, you —

- ◯ **A.** think very highly of that person.
- ◯ **B.** race against them in a game.
- ◯ **C.** forget who they are.
- ◯ **D.** measure how tall they are.

CRITICAL VOCABULARY · THE INVENTOR'S SECRET

Q5 L.5.4

locomotives

Young Thomas was curious about **locomotives**. **Locomotives** are —

- ◯ **A.** the cars where train passengers sit.
- ◯ **B.** the metal tracks a train rides along.
- ◯ **C.** the railway workers who repair trains.
- ◯ **D.** the engines that make a train go forward.

Q6 L.5.5 TRAP

chugged

Henry's engines **chugged** and purred. If a machine **chugged** along, it —

- ◯ **A.** moved silently and very fast.
- ◯ **B.** moved slowly and noisily.
- ◯ **C.** stayed completely still.
- ◯ **D.** floated up into the air.

Q7 L.5.4

gadgets

Thomas dreamed of electric **gadgets** to make life easier. **Gadgets** are —

- ◯ **A.** large, heavy machines too big to carry.
- ◯ **B.** plain containers with no moving parts.
- ◯ **C.** simple tools that use no power at all.
- ◯ **D.** small, specialized machines or electronic devices.

Q8 L.5.4

phonograph

Thomas talked into his new **phonograph**. A **phonograph** is —

- ◯ **A.** a machine that plays recorded music or sound.
- ◯ **B.** a machine that sends messages by wire.
- ◯ **C.** a machine that takes moving pictures.
- ◯ **D.** a machine that prints copies of writing.

Q9 L.5.4

sputtered

Henry's contraption **sputtered** along for forty feet. If something **sputtered**, it —

- ◯ **A.** ran perfectly smoothly the whole time.
- ◯ **B.** worked in a rough, uneven way and made popping noises.
- ◯ **C.** turned a bright color.
- ◯ **D.** floated quietly on the water.

CRITICAL VOCABULARY · THE INVENTOR'S SECRET (CONTINUED)

Q10 L.5.5 TRAP

flop

Henry's design was a **flop**! Something that is a **flop** is —

- ◯ **A.** a complete failure.
- ◯ **B.** a huge success.
- ◯ **C.** a small repair.
- ◯ **D.** a careful plan.

Q11 L.5.4

incandescent

Thomas created an **incandescent** light bulb that burned all night. Something **incandescent** —

- ◯ **A.** makes a loud sound.
- ◯ **B.** is very heavy to lift.
- ◯ **C.** gives off a lot of light.
- ◯ **D.** is frozen and cold.

Q12 L.5.4

cylinder

The engine had a four-stroke **cylinder** that burned gas. A **cylinder** —

- **A.** is a flat square sheet of metal.
- **B.** is a tall steel tower.
- **C.** is a kind of light bulb.
- **D.** has circular ends and straight sides and, in an engine, takes in gas to make other parts move.

Q13 L.5.5 TRAP

patents

Thomas applied for **patents** for his inventions. **Patents** are —

- **A.** legal documents that mean no one else is allowed to make or sell an invention.
- **B.** permissions that let everyone copy an invention freely.
- **C.** prizes given for winning a race.
- **D.** drawings of faraway places.

CRITICAL VOCABULARY · WINDS OF HOPE

Q14 L.5.4

irrigate

A windmill could pump water to **irrigate** the maize fields. To **irrigate** crops is to —

- **A.** cut them down and store them.
- **B.** sell them at the market.
- **C.** supply them with water through pipes, sprinklers, or streams.
- **D.** count how many there are.

Q15 L.5.4

inspector

A school **inspector** saw the windmill. An **inspector** is someone who —

- **A.** reviews or examines something carefully.
- **B.** drives a school bus.
- **C.** builds tall towers.
- **D.** plants crops in a field.

Q16 L.5.5 TRAP

photographed

William was interviewed and **photographed** for the newspapers. If something was **photographed**, its photo was —

- ◯ **A.** erased so no one could see it.
- ◯ **B.** drawn by hand with a pencil.
- ◯ **C.** spoken aloud on the radio.
- ◯ **D.** recorded on film or as a computer file.

Q17 L.5.4

prestigious

William spoke at the **prestigious** TED conference. Something **prestigious** is —

- ◯ **A.** impressive and important.
- ◯ **B.** tiny and easy to miss.
- ◯ **C.** broken and useless.
- ◯ **D.** far away and empty.

Q18 L.5.4

auditorium

William received an ovation from the **auditorium** of inventors. An **auditorium** is —

- ◯ **A.** a small handheld device.
- ◯ **B.** a large room where an audience gathers for a presentation or performance.
- ◯ **C.** a machine that makes electricity.
- ◯ **D.** a dry field during a drought.

CRITICAL VOCABULARY · WINDS OF HOPE (CONTINUED)

Q19 L.5.4

impoverished

William brings electricity to **impoverished** villages. To be **impoverished** is to be —

- ◯ **A.** very crowded.
- ◯ **B.** far underground.
- ◯ **C.** brand new.
- ◯ **D.** poor.

Q20 L.5.4

maneuver

The heavy wheelchair was hard to **maneuver**. To **maneuver** something is to —

- **A.** hold it firmly in one place.
- **B.** leave it where it is.
- **C.** move it skillfully into position.
- **D.** take it apart piece by piece.

Q21 L.5.4

specialized

The company **specialized** in meeting the needs of athletes. If a company **specialized** in something, it —

- **A.** sold a little of everything.
- **B.** provided a specific type of product.
- **C.** gave its products away for free.
- **D.** closed down after one year.

Q22 L.5.5 TRAP

elite

Wheelchair athletes can compete at an **elite** level. **Elite** members of a group are those who are —

- **A.** the newest and least practiced.
- **B.** the most tired in the group.
- **C.** the youngest in the group.
- **D.** the best or most skilled.

Q23 L.5.4

objective

Designers redesigned wheelchairs to meet athletes' **objectives**. An **objective** is —

- **A.** a sudden problem.
- **B.** a finished result.
- **C.** a goal someone is trying to reach.
- **D.** a careful set of rules.

Q24 L.5.5 TRAP

traditional

A sports wheelchair is lighter than a **traditional** wheelchair. Something **traditional** has been —

- ◯ **A.** made or done in a certain way for a very long time.
- ◯ **B.** invented just last week.
- ◯ **C.** kept secret from everyone.
- ◯ **D.** built only once and then thrown away.

CRITICAL VOCABULARY · CAPTAIN ARSENIO

Q25 L.5.4

passionate

Flight has inspired a **passionate** urge in many people. To have a **passionate** feeling is to have —

- ◯ **A.** no feelings at all about it.
- ◯ **B.** strong emotions about it.
- ◯ **C.** a sleepy feeling about it.
- ◯ **D.** a bored feeling about it.

CRITICAL VOCABULARY · CAPTAIN ARSENIO (CONTINUED)

Q26 L.5.4

impulse

A passionate **impulse** pushed people to try to fly. An **impulse** is —

- ◯ **A.** a kind of flying machine.
- ◯ **B.** a careful written plan.
- ◯ **C.** a finished invention.
- ◯ **D.** the desire to do something.

Q27 L.5.4

contribution

Each pioneer made a different **contribution** to flight. A person who helps to make something has made a **contribution** —

- ◯ **A.** by refusing to help at all.
- ◯ **B.** by watching from far away.
- ◯ **C.** to that work.
- ◯ **D.** only after it is finished.

Q28 L.5.5 TRAP

distinguished

Captain Arsenio entered the **distinguished** pages of aviation history. A **distinguished** group is —

- ◯ **A.** looked down on and ignored.
- ◯ **B.** known and respected for its excellence.
- ◯ **C.** hidden away from everyone.
- ◯ **D.** ordinary and easily forgotten.

Q29 L.5.5 TRAP

eccentric

Captain Arsenio built a series of **eccentric** projects. Someone who is **eccentric** is —

- ◯ **A.** odd.
- ◯ **B.** very ordinary.
- ◯ **C.** extremely tall.
- ◯ **D.** always silent.

Q30 L.5.4

circumstances

Arsenio's diary was found under mysterious **circumstances**. The **circumstances** of an event are —

- ◯ **A.** the people who watched it on a screen.
- ◯ **B.** the prizes given out afterward.
- ◯ **C.** the way it happened or the causes of it.
- ◯ **D.** the pictures drawn about it.

Q31 L.5.5 TRAP

evidently

The text says the Motocanary **evidently** demanded a lot of work. If something happened **evidently**, it happened —

- ◯ **A.** for a hidden, secret reason.
- ◯ **B.** for no reason that anyone could find.
- ◯ **C.** by complete accident.
- ◯ **D.** for an obvious reason.

Q32 L.5.4

acceleration

The **acceleration** of the runner would lift the machine. **Acceleration** is the act of —

- **A.** stopping completely.
- **B.** turning around.
- **C.** moving faster.
- **D.** resting quietly.

Q33 L.5.5 TRAP

prototype

Captain Arsenio tested a **prototype** before the final version. A **prototype** is —

- **A.** the finished, polished product ready to sell.
- **B.** a rough model created to test something before creating it in its final form.
- **C.** a legal document about an invention.
- **D.** a prize for the best inventor.

Q34 L.5.4

conceived

No one knows how Arsenio **conceived** of his contraption. If you **conceived** something, you —

- **A.** thought of the idea to create it.
- **B.** copied it from a friend.
- **C.** threw the idea away.
- **D.** finished building it.

B Worksheet B — Reading Comprehension

Module 1 · Inventors at Work · Grade 5 · 32 Items

CCSS RI.5.1 · RI.5.2 · RI.5.3 · RI.5.5 · RI.5.6 · RI.5.8 · RL.5.1 · RL.5.2 · RL.5.3 · RL.5.6

NAME _____ DATE _____

Directions. Use what you remember from each selection to answer the questions. Fill in the circle next to the BEST answer.

Selection 1 · The Inventor's Secret

NARRATIVE NONFICTION · SUZANNE SLADE · 8 QUESTIONS

Q1 RI.5.2

What is "The Inventor's Secret" mostly about?

- ◯ **A.** How a heavy steel wheelchair was redesigned into a light sports chair so athletes could compete.
- ◯ **B.** How a boy in Malawi built a windmill from junkyard scrap to power his village during a drought.
- ◯ **C.** How two curious boys, Thomas Edison and Henry Ford, became great inventors through hard work and not giving up.
- ◯ **D.** How an eccentric captain kept building flying machines that crashed in comical ways.

Q2 RI.5.5

How does the author organize most of this text?

- ◯ **A.** By alternating between Thomas and Henry to compare their lives over time.
- ◯ **B.** As a list of recipes to follow in order.
- ◯ **C.** As a single letter from one inventor to another.
- ◯ **D.** By listing the parts of an engine from largest to smallest.

Q3 RI.5.3

According to the text, how were Thomas and Henry ALIKE as boys?

- ◯ **A.** They both grew up in the same town.
- ◯ **B.** They were born in the same year.
- ◯ **C.** They were both very curious and loved to experiment to see how things worked.
- ◯ **D.** They both built a working car as children.

Q4 RI.5.3

What happened when Henry strapped his homemade engine to an old mowing machine?

- ◯ **A.** It won a prize at a fair.
- ◯ **B.** His contraption sputtered along for forty feet, then collapsed.
- ◯ **C.** It turned into a phonograph.
- ◯ **D.** It ran for many miles without stopping.

Q5 RI.5.6

From whose point of view is the story told?

- ◯ **A.** Thomas Edison, telling his own story with the word "I."
- ◯ **B.** A narrator who is outside the story and uses words like *he*, telling about both Thomas and Henry.
- ◯ **C.** Henry Ford, telling his own story with the word "I."
- ◯ **D.** A talking machine that watched it all happen.

Q6 RI.5.1

Why does Henry keep wondering, "What's his secret?"

- ◯ **A.** He badly wants to figure out how Thomas keeps making such successful inventions.
- ◯ **B.** He wants to know where Thomas lives.
- ◯ **C.** He is trying to find a lost menu.
- ◯ **D.** He wants to learn how to ride a horse.

Q7 RI.5.8

The author shows that great inventing takes persistence. Which detail BEST supports this point?

- ◯ **A.** Even after his designs failed, Henry kept working long nights until he built a car people could afford.
- ◯ **B.** Henry was born sixteen years after Thomas.
- ◯ **C.** Thomas sold papers on a train as a newsboy.
- ◯ **D.** Thomas tied wires to his cats' tails.

Q8 RI.5.2

What is the "secret" that Henry realizes he had known all along?

- ◯ **A.** A secret invention that Thomas kept hidden away in his workshop.
- ◯ **B.** That the key to success is to keep working hard and never give up.
- ◯ **C.** A special tool that only Thomas owned and never shared with Henry.
- ◯ **D.** A map showing where to find gold to pay for the experiments.

Selection 2 · Winds of Hope

NARRATIVE NONFICTION / MAGAZINE ARTICLE · KATY DUFFIELD · 8 QUESTIONS

Q9 RI.5.2

What is "Winds of Hope" mostly about?

- **A.** How an eccentric captain built flying machines that crashed again and again.
- **B.** How the curious boys Thomas Edison and Henry Ford grew up to be inventors.
- **C.** How wheelchairs were redesigned so disabled athletes could train and compete.
- **D.** How William Kamkwamba built a windmill from scrap to bring electricity and water to his village.

Q10 RI.5.3

What circumstance made William's family suffer at the start of the article?

- **A.** A flood washed away their home.
- **B.** Their cows ran away in the night.
- **C.** A storm knocked down their windmill.
- **D.** A drought meant their maize would not grow, and they had little to eat.

Q11 RI.5.3

Why did William's education stop at eighth grade?

- **A.** He decided he did not want to learn anymore.
- **B.** Because of the drought, his family had no money to pay his school fees.
- **C.** His family moved to another country.
- **D.** The school closed down for good.

Q12 RI.5.5

How is most of this magazine article organized?

- **A.** As a poem with rhyming lines.
- **B.** By comparing two made-up characters.
- **C.** In chronological order, telling William's true story year by year.
- **D.** As a list of facts with no order.

Q13 RI.5.3

How did William find the parts to build his windmill?

- **A.** He ordered the parts from another country.
- **B.** He bought a windmill kit from a store.
- **C.** A factory gave him brand-new parts.
- **D.** He dug through a junkyard for scrap metal, old tractor parts, and a bicycle frame.

Q14 RI.5.1

What does the article suggest about William's character?

- ◯ **A.** He gives up quickly when things are hard.
- ◯ **B.** He cares only about becoming famous.
- ◯ **C.** He is curious, creative, and determined, learning on his own even without school.
- ◯ **D.** He waits for others to solve his problems.

Q15 RI.5.8

The article shows that William's invention changed his community. Which detail BEST supports this?

- ◯ **A.** William liked to read science books.
- ◯ **B.** The library had only three shelves.
- ◯ **C.** William was fourteen years old during the drought.
- ◯ **D.** His windmill powered lights and radios in his house and even recharged village cell phones.

Q16 RI.5.2

Which sentence BEST states a central idea of the article?

- ◯ **A.** A windmill is the only machine that can ever make electricity for a village.
- ◯ **B.** With creativity and persistence, even a boy with little schooling can invent something that transforms his community.
- ◯ **C.** A drought always lasts for many years and there is nothing anyone can do.
- ◯ **D.** Reading science books in a library is a waste of a young person's time.

Selection 3 · Wheelchair Sports: Hang Glider to Wheeler-Dealer

INFORMATIONAL TEXT · SIMON SHAPIRO · 8 QUESTIONS

Q17 RI.5.2

What is "Wheelchair Sports" mostly about?

- ◯ **A.** How the inventors Edison and Ford built the phonograph and the first cars.
- ◯ **B.** How wheelchairs and wheelchair sports have been invented and improved so athletes can compete.
- ◯ **C.** How a boy in Malawi built a windmill from scrap during a long drought.
- ◯ **D.** How an eccentric captain tried to fly with machines powered by canaries.

Q18 RI.5.3

What led Marilyn Hamilton to imagine a new kind of wheelchair?

- ◯ **A.** She read about it in a science book.
- ◯ **B.** She wanted to win a prize for inventing.
- ◯ **C.** A company asked her to design one.
- ◯ **D.** Her heavy, hard-to-maneuver chair frustrated her when she tried wheelchair tennis.

Q19 RI.5.3

How was the new "Quickie" wheelchair different from Marilyn's old one?

- ◯ **A.** It was made of heavier steel.
- ◯ **B.** It could not be moved at all.
- ◯ **C.** It used a light aluminum frame, so it was strong but about half the weight and easy to move.
- ◯ **D.** It had no wheels.

Q20 RI.5.5

In the "Why They're Better" section, how does the author explain why a lighter wheelchair helps athletes?

- ◯ **A.** By telling a funny made-up story.
- ◯ **B.** By listing the athletes' names.
- ◯ **C.** By describing the colors of the wheels.
- ◯ **D.** By explaining cause and effect, using ideas about force and center of mass.

Q21 RI.5.6

Why does the author include the diagrams of the wheelchairs?

- ◯ **A.** To list the price of each chair.
- ◯ **B.** To show how to cook a meal.
- ◯ **C.** To help readers understand a science idea — how center of mass affects tipping — that is hard to grasp from words alone.
- ◯ **D.** To tell a joke about athletes.

Q22 RI.5.3

According to the "Guttmann's Great Idea" section, how was Dr. Guttmann's idea different from what had been done before?

- ◯ **A.** He told patients to stay in bed and rest.
- ◯ **B.** He sent the patients home right away.
- ◯ **C.** He used sports to get patients active instead of leaving them lying in bed doing nothing.
- ◯ **D.** He refused to treat the patients at all.

Q23 RI.5.8

The author argues that inventing better equipment lets athletes compete. Which detail BEST supports this?

- **A.** Marilyn Hamilton went hang gliding in California.
- **B.** The lighter Quickie chair let athletes move faster, stop faster, and change direction more easily.
- **C.** Sir Isaac Newton lived long ago.
- **D.** The first wheelchair was blue, not yellow.

Q24 RI.5.2

Which sentence BEST states a central idea of the text?

- **A.** Wheelchairs should never be changed from their original heavy design.
- **B.** Inventing and adapting specialized equipment lets athletes with disabilities compete and excel.
- **C.** Hang gliding is the safest sport for people who use wheelchairs.
- **D.** Only doctors, not athletes, are allowed to invent new sports equipment.

Selection 4 · Captain Arsenio

SCIENCE FICTION / FANTASY · PABLO BERNASCONI · 8 QUESTIONS

Q25 RL.5.2

What is "Captain Arsenio" mostly about?

- **A.** An athlete who redesigns a heavy wheelchair into a lighter sports chair.
- **B.** A determined boy who builds a windmill from scrap during a drought.
- **C.** Two real inventors who build the first cars and the incandescent light bulb.
- **D.** An eccentric character who keeps trying — and comically failing — to build a flying machine.

Q26 RL.5.3

What do Captain Arsenio's actions show about his character?

- **A.** He is confident and determined but reckless, never thinking through the dangers.
- **B.** He carefully plans every detail before acting.
- **C.** He gives up after his very first try.
- **D.** He is shy and never tries anything new.

Q27 RL.5.6

How does the narrator's view of events differ from Captain Arsenio's?

- **A.** The narrator believes every experiment is a success.
- **B.** The narrator and the captain agree on everything.
- **C.** The captain is sure he will succeed, but the narrator points out the obvious problems and failures.
- **D.** The narrator refuses to describe any of the inventions.

Q28 RL.5.3

How does the Motocanary project end?

- A. It flies only a few feet before crashing into a tree, leaving the captain stuck for two days.
- B. Captain Arsenio decides not to test it.
- C. The canaries refuse to move.
- D. The machine flies safely for miles.

Q29 RL.5.1

The captain calls a huge crash "a small problem." What does this tell you about him?

- A. He does not like to admit how badly things go or how wrong he might be.
- B. He is a careful, honest reporter of facts.
- C. He has stopped caring about flying.
- D. He never writes anything in his diary.

Q30 RL.5.3

How are the Motocanary and the Flying Runner projects ALIKE?

- A. Both work perfectly the first time.
- B. Both are built by Henry Ford.
- C. Both use the same canaries.
- D. Both seem very dangerous and risky, yet the captain is sure each one will work.

Q31 RL.5.6

What is the author's main purpose in writing this story?

- A. To entertain readers with a funny story about a made-up inventor and his silly machines.
- B. To warn readers never to invent anything.
- C. To teach readers exactly how to build a real airplane.
- D. To report the true history of flight.

Q32 RL.5.2 TRAP

What is the main THEME, or lesson, of "Captain Arsenio"?

- A. Invention is driven by passion and imagination, and even comical failures are part of the inventive adventure.
- B. Captain Arsenio's diary had ninety pages full of doodles, notes, and technical writings.
- C. The story names earlier flight dreamers, from Icarus to the Wright brothers.
- D. The Motocanary tried to fly using many canaries tied together with a rope.

Worksheet C — IB Critical Thinking

Module 1 · Inventors at Work · Grade 5

IB PYP Key Concepts · DOK 2 – 4

NAME _____ DATE _____

Directions. Use what you learned from each selection to think more deeply about why people invent. Each question helps you grow as an IB learner.

Q1 Perseverance DOK 3

In **"The Inventor's Secret,"** Thomas and Henry both failed many times before they succeeded. Show one FAILURE and one later SUCCESS, then explain "the secret."

HOW INVENTORS KEEP TRYING · THE INVENTOR'S SECRET

A FAILURE	A LATER SUCCESS	THE SECRET

The same question — "What's his secret?" — comes up again and again. Why do you think the author repeats it? What is the secret Henry finally understands? Use the word *persistence* in your answer.

Q2 Circumstances DOK 3

In **"Winds of Hope,"** hard circumstances pushed William to invent. Show WHAT problem he faced, then explain HOW he solved it.

A PROBLEM PUSHES AN INVENTION · WINDS OF HOPE

THE PROBLEM	WHAT HE DID	THE RESULT
what circumstances did William face?	*how did he build the windmill?*	*what did the windmill do?*

William had little money and could not stay in school. Why is it amazing that he still invented a working windmill? Explain using the word *circumstances*.

Q3 Innovation DOK 3

In **"Wheelchair Sports,"** a traditional wheelchair was redesigned into a sports wheelchair. Compare the TRADITIONAL chair with the SPORTS chair, then explain the lesson.

REDESIGNING TO SOLVE A PROBLEM · WHEELCHAIR SPORTS

TRADITIONAL WHEELCHAIR	SPORTS "QUICKIE" WHEELCHAIR

Marilyn Hamilton used hang-glider technology to redesign the wheelchair. Why does inventing better, specialized equipment help athletes excel? Explain.

Q4 Creativity DOK 3

In **"Captain Arsenio,"** the captain's inventions all fail, yet he never stops. Show how he feels at the START of a project and how it ENDS, then explain what the story teaches.

PASSION THROUGH FAILURE · CAPTAIN ARSENIO

AT THE START OF A PROJECT	HOW THE PROJECT ENDS

Captain Arsenio begins each project sure that "It cannot fail!" Why does the author make us laugh at his failures and yet admire his passion? Use the word *imagination* in your answer.

Q5 Inquiry DOK 4

Essential Question: What kinds of circumstances push people to create new inventions? Use ideas from TWO of the selections in this module to answer.

TWO-SELECTION SYNTHESIS

SELECTION 1 · TITLE

What circumstance pushed someone to invent in this selection?

SELECTION 2 · TITLE

What circumstance pushed someone to invent in this selection?

MY BIG IDEA

Worksheet D — MAP Practice

Module 1 · Inventors at Work · Grade 5

MAP RIT 205 → 225 · 4 Mini-Passages × 5 Questions

NAME _____ DATE _____

Directions. Read each short passage carefully. Then answer the 5 questions that follow. Each passage is NEW — not from the selections in this workbook.

Passage 1 · The Lady of the Lamp's Big Idea

More than a hundred years ago, hospitals were often dirty and dangerous places. Many patients did not get better; instead, they grew sicker. A nurse named Florence Nightingale wanted to change this. During a war, she was sent to care for wounded soldiers in a crowded army hospital. The rooms were filthy, and many men died not from their wounds but from illnesses that spread in the dirt. Nightingale believed that clean rooms, fresh air, and good food could help the soldiers heal. She set to work scrubbing the wards, opening windows, and feeding the men proper meals. To prove that her ideas worked, she did something clever. She carefully counted how many soldiers died each month and drew her numbers as colorful charts shaped like circles. The pictures made it easy for everyone to see that fewer men died once the hospital was clean. Leaders who saw her charts were convinced, and hospitals everywhere began to change. Because of Nightingale's careful work and her new way of showing facts with pictures, nursing became a respected job, and countless lives were saved. Her circumstances pushed her to invent better ways of caring for the sick.

195 WORDS · LEXILE 720–800L · RIT 205–211

Q1 RIT 205 | Lexile 720L | Key Detail

Where was Florence Nightingale sent to care for soldiers?

- ◯ **A.** A farm in the countryside.
- ◯ **B.** A school in a quiet town.
- ◯ **C.** A factory that made charts.
- ◯ **D.** A crowded army hospital during a war.

Q2 RIT 207 | Lexile 740L | Cause & Effect

Why were many soldiers dying in the hospital?

- ◯ **A.** Illnesses spread in the dirt, not just because of their wounds.
- ◯ **B.** They were sent home too early.
- ◯ **C.** There was too much fresh air.
- ◯ **D.** They ate too many proper meals.

Q3 RIT 208 · Lexile 760L · Key Detail

What clever thing did Nightingale do to prove her ideas worked?

- ◯ **A.** She wrote a long song about the war.
- ◯ **B.** She counted how many soldiers died each month and drew the numbers as colorful charts.
- ◯ **C.** She built a new hospital by herself.
- ◯ **D.** She moved all the patients to a farm.

Q4 RIT 210 · Lexile 780L · Vocabulary

The passage says leaders "were **convinced**" by her charts. The word **convinced** means they were —

- ◯ **A.** made angry.
- ◯ **B.** put to sleep.
- ◯ **C.** left confused.
- ◯ **D.** made to believe she was right.

Q5 RIT 211 · Lexile 800L · Main Idea

What is this passage MOSTLY about?

- ◯ **A.** Why wars in faraway countries usually last for many long years.
- ◯ **B.** How to draw numbers as colorful circle-shaped charts step by step.
- ◯ **C.** How Florence Nightingale's circumstances pushed her to invent better ways to care for the sick and to show facts with pictures.
- ◯ **D.** How soldiers were trained to march into battle during the war.

Passage 2 · The Accidental Invention

Not every invention comes from a careful plan. Some of the most useful objects in our homes began as happy accidents. Take the sticky note, the little square of paper that clings to a desk and then peels off without leaving a mark. It exists because of a mistake. In 1968, a scientist named Spencer Silver was trying to create a very strong glue. Instead, he made a glue that was surprisingly weak. It stuck things together gently, but it could be pulled apart again and again. Silver thought his weak glue might be useful, but for years no one knew what to do with it. Then another scientist, Art Fry, had a problem. He sang in a choir, and the paper bookmarks kept slipping out of his hymn book. He remembered Silver's weak glue and dabbed a little on the back of a bookmark. Now the bookmark stayed in place but did not tear the page when removed. The two scientists had turned a failed glue into a brand-new product. Today, people use sticky notes all over the world. The lesson is clear: sometimes a mistake, looked at in a new way, can lead to a great invention.

199 WORDS · LEXILE 760–840L · RIT 211–217

Q6 RIT 211 · Lexile 760L · Key Detail

What was Spencer Silver trying to make in 1968?

- ◯ **A.** A new kind of paper.
- ◯ **B.** A song for his choir.
- ◯ **C.** A bookmark for a hymn book.
- ◯ **D.** A very strong glue.

Q7 RIT 213 · Lexile 790L · Key Detail

What problem did Art Fry have?

- ◯ **A.** His glue was too strong.
- ◯ **B.** The paper bookmarks kept slipping out of his hymn book.
- ◯ **C.** He could not find any paper.
- ◯ **D.** He forgot the words to the songs.

Q8 RIT 214 · Lexile 810L · Cause & Effect

Why was the weak glue perfect for a bookmark?

- ◯ **A.** It held the bookmark in place but could be removed without tearing the page.
- ◯ **B.** It made the paper heavier.
- ◯ **C.** It glowed in the dark.
- ◯ **D.** It was very colorful.

Q9 RIT 215 · Lexile 820L · Vocabulary

The passage says the sticky note "**clings** to a desk." In this sentence, **clings** means —

- ◯ **A.** falls straight down.
- ◯ **B.** holds on tightly.
- ◯ **C.** floats away.
- ◯ **D.** melts slowly.

Q10 RIT 217 · Lexile 840L · Main Idea

What is the main idea of "The Accidental Invention"?

- ◯ **A.** Choirs need better bookmarks that will not slip out of a hymn book.
- ◯ **B.** Strong glue is always more useful than a glue that is weak.
- ◯ **C.** Sometimes a mistake, looked at in a new way, can lead to a great invention.
- ◯ **D.** Scientists should never make mistakes when they are doing their work.

Passage 3 · The Girl Who Invented a Sport

Most sports have been around for a very long time, but a few are brand new. One of the newest is a fast, exciting game called "Newcomb pickle," invented by a twelve-year-old named Priya. Priya loved playing games at recess, but her younger brother used a wheelchair and could not join in the games that needed running. This bothered her. She believed that a good game should let everyone play together, no matter how they moved. So Priya set out to invent a sport of her own. First, she made a list of rules from games she already liked. Then she changed the rules so that no one had to run far. Players could roll, walk, or wheel to reach the ball. The court was small, and the ball was soft and light, so it was easy to catch and throw. After many tries, Priya tested her game with her class. Some rules did not work, so she fixed them, just as an inventor improves a machine. Soon her whole school was playing. Priya's invention proves that a problem you care about, plus a creative idea, can lead to something new that brings people together.

195 WORDS · LEXILE 800–880L · RIT 217–222

Q11 RIT 217 · Lexile 800L · Key Detail

Who invented the new game in this passage?

- ◯ **A.** A famous athlete.
- ◯ **B.** A twelve-year-old named Priya.
- ◯ **C.** Priya's younger brother.
- ◯ **D.** A gym teacher.

Q12 RIT 218 · Lexile 820L · Cause & Effect

What circumstance pushed Priya to invent her sport?

- ◯ **A.** She wanted to win a prize.
- ◯ **B.** Her teacher gave her homework.
- ◯ **C.** Her younger brother, who used a wheelchair, could not join games that needed running.
- ◯ **D.** She was bored at home.

Q13 RIT 219 · Lexile 840L · Sequence

What did Priya do FIRST when she set out to invent her game?

- ◯ **A.** She made a list of rules from games she already liked.
- ◯ **B.** She taught the game to her whole school.
- ◯ **C.** She bought a soft, light ball.
- ◯ **D.** She tested the game with her class.

Q14 RIT 220 · Lexile 860L · Inference

When the passage says Priya "fixed them, just as an inventor improves a machine," it shows that —

- ◯ **A.** Priya built a real machine.
- ◯ **B.** the game needed no changes at all.
- ◯ **C.** Priya gave up on the game.
- ◯ **D.** inventing often means testing, finding problems, and improving an idea.

Q15 RIT 222 · Lexile 880L · Main Idea

What is the BIG IDEA of "The Girl Who Invented a Sport"?

- A. Old sports that have been around a long time are better than new ones.
- B. Recess at school should be made much longer for everyone.
- C. A problem you care about, plus a creative idea, can lead to something new that brings people together.
- D. Soft, light balls are always easier to throw and catch than hard balls.

Passage 4 · The Inventor in the Workshop

The workshop smelled of sawdust and oil. Maya pushed open the door and stepped inside, where her grandfather sat hunched over his workbench, surrounded by gears, springs, and bits of wire. He had spent his whole life fixing clocks, but lately he had a new dream: to build a music box that played a tune he remembered from his childhood. "It still doesn't work," he sighed, pointing at a tangle of metal. Maya pulled up a stool and watched him wind a tiny spring. The box gave a single weak note, then fell silent. "Maybe the spring is too small," Maya suggested. Her grandfather raised an eyebrow, then smiled. Together they tried a larger spring. This time the box chimed three clear notes before stopping. "Closer," he said. They worked all afternoon, testing one idea after another. Each failure taught them something new. By evening, the little box played the whole tune, soft and bright. Maya's grandfather closed his eyes and listened, and she saw a tear at the corner of his eye. In that moment, Maya understood that inventing was not about getting it right the first time. It was about caring enough to keep trying, together, until the idea came to life.

204 WORDS · LEXILE 860–950L · RIT 219–225

Q16 RIT 219 · Lexile 860L · Key Detail

What new dream did Maya's grandfather have?

- A. To open a sawmill.
- B. To sell his old clocks.
- C. To build a music box that played a tune from his childhood.
- D. To teach Maya how to fix wire.

Q17 RIT 220 · Lexile 880L · Cause & Effect

What happened after they tried a larger spring?

- A. The box made no sound at all.
- B. The box broke into pieces.
- C. The box played the whole tune at once.
- D. The box chimed three clear notes before stopping.

Q18 RIT 221 | Lexile 900L | Vocabulary

The passage says the grandfather sat "**hunched** over his workbench." The word **hunched** means he —

- ◯ **A.** bent his back and leaned forward.
- ◯ **B.** stood up straight and tall.
- ◯ **C.** ran quickly around the room.
- ◯ **D.** lay down to sleep.

Q19 RIT 223 | Lexile 920L | Inference

Why does the grandfather have a tear at the corner of his eye at the end?

- ◯ **A.** He is upset that the box still does not work.
- ◯ **B.** He is moved with joy that, after many tries, the box finally plays the tune he remembered.
- ◯ **C.** He is tired and wants to go home.
- ◯ **D.** The sawdust is bothering his eyes.

Q20 RIT 225 | Lexile 950L | Theme

What is the theme, or lesson, of this story?

- ◯ **A.** Inventing is not about getting it right the first time; it is about caring enough to keep trying until the idea comes to life.
- ◯ **B.** Music boxes are old-fashioned things that are not worth building anymore.
- ◯ **C.** Grandparents always know the answer to every problem right away.
- ◯ **D.** Clocks are much harder to fix than a small wind-up music box.